

# Environmental Engineering Management

50 Integrated Questions on Carbon Footprints, Energy  
Intensity,  
Water Intensity, and Air Pollution Economics

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**Course Material for BSc in Electrical and Electronics  
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#### Key Data for Tanzania (Use where applicable)

- TANESCO grid emission factor (2023): 0.531 kgCO<sub>2</sub>e/kWh
- Solar PV lifecycle EF: 0.04 kgCO<sub>2</sub>e/kWh
- Wind power lifecycle EF: 0.02 kgCO<sub>2</sub>e/kWh
- Hydropower lifecycle EF: 0.01 kgCO<sub>2</sub>e/kWh
- Geothermal lifecycle EF (assumed): 0.03 kgCO<sub>2</sub>e/kWh
- Diesel generator EF: 0.85 kgCO<sub>2</sub>e/kWh (combustion only)
- Energy intensity of electricity (Tanzania average): 3.5 MJ/USD (illustrative)
- Water intensity for solar PV manufacturing: 0.6 L/W<sub>p</sub>
- Water intensity for electronics (e.g., inverters): 100 L/kg
- PM<sub>2.5</sub> mortality cost: 1.5 million TZS per  $\mu\text{g}/\text{m}^3$  exposure per person
- Economic cost of air pollution from diesel gensets: 5000 TZS/kWh

## Part 1: Carbon Footprint of Electricity Production and Consumption (Questions 1–20)

### Question

1. A residential building in Dar es Salaam consumes 450 kWh per month from the TANESCO grid. Calculate its annual carbon footprint in kgCO<sub>2</sub>e. If the household switches to a solar PV system covering 80% of its consumption, what is the annual reduction in kgCO<sub>2</sub>e? (Use grid EF = 0.531 kgCO<sub>2</sub>e/kWh).

### Question

2. A small off-grid clinic uses a diesel generator with an emission factor of 0.85 kgCO<sub>2</sub>e/kWh. The generator runs 6 hours daily, supplying 5 kW average load. Determine the daily and annual CO<sub>2</sub>e emissions. Compare with the emissions if the same energy were supplied by a solar + battery system (use solar lifecycle EF).

### Question

3. A wind farm in Tanzania generates 120 000 MWh per year. Its lifecycle emission factor is 0.02 kgCO<sub>2</sub>e/kWh. Calculate the farm's annual footprint. If this wind power displaces grid electricity (EF = 0.531), what is the annual avoided emissions (tonnes CO<sub>2</sub>e)?

#### Question

4. A hydropower plant produces 50 000 MWh/year ( $EF = 0.01$ ). A natural gas plant with  $EF = 0.45$  kgCO<sub>2</sub>e/kWh would otherwise produce the same energy. Compute the annual emission savings from using hydropower.

#### Question

5. A factory consumes 200 000 kWh/month from the grid. It plans to install a 150 kWp solar PV system. Estimate the monthly solar generation (use 5 peak sun hours/day, 0.8 performance ratio). Then compute the new monthly carbon footprint (grid + solar) and the reduction compared to grid-only (grid  $EF = 0.531$ ).

#### Question

6. A geothermal plant (lifecycle  $EF = 0.03$  kgCO<sub>2</sub>e/kWh) generates 80 000 MWh/year. Calculate its footprint and the emissions avoided if it replaces grid electricity (grid  $EF = 0.531$ ).

#### Question

7. A Tanzanian university campus has an annual electricity use of 2.5 GWh. It plans to meet 40% from wind ( $EF = 0.02$ ) and the rest from grid. What is the total annual carbon footprint? How many tonnes are saved compared to using 100% grid? (Grid  $EF = 0.531$ ).

#### Question

8. A remote mine uses a 2 MW diesel generator running 20 hours/day. Compute the daily CO<sub>2</sub>e emissions ( $EF_{\text{diesel}} = 0.85$ ). If a solar-wind hybrid with battery (lifecycle  $EF = 0.03$  per kWh) replaces 75% of diesel generation, what is the daily emission reduction?

#### Question

9. Compare the lifecycle carbon footprints of three technologies – solar PV, wind, and hydro – per MWh generated. If a country replaces 500 GWh/year of coal ( $EF = 1.0$ ) with an equal mix of these three, calculate the total avoided emissions.

#### Question

10. A commercial building uses 15 000 kWh/month grid electricity ( $EF = 0.531$ ). After installing a 50 kWp solar system, its net grid consumption drops to 60% of original. Calculate the annual carbon footprint before and after, assuming solar lifecycle  $EF$ .

#### Question

11. A hospital's annual electricity demand is 8 000 MWh. It sources 30% from on-site solar ( $EF = 0.04$ ) and 20% from a wind PPA ( $EF = 0.02$ ), the rest from grid ( $EF = 0.531$ ). Compute total annual CO<sub>2</sub>e footprint.

### Question

**12.** A data centre consumes 10 GWh/year. The operator considers two options: (A) 100% grid (EF 0.531), (B) 60% solar+40% wind. Calculate the footprint reduction (tonnes) when moving from A to B.

### Question

**13.** A Tanzanian household replaces an old refrigerator (200W, running 10h/day) with an energy-efficient model (120W). Grid EF is 0.531. How many kgCO<sub>2</sub>e are saved annually from this efficiency measure alone?

### Question

**14.** An industrial park has a peak demand of 5 MW. During peak hours (6h/day), 70% of the energy comes from a diesel genset (EF 0.85) and 30% from grid (EF 0.531). Calculate the daily CO<sub>2</sub>e from the peak period.

### Question

**15.** A solar module manufacturing plant consumes 2 GWh/year from the grid (EF 0.531). The plant wishes to become carbon-neutral by installing a rooftop solar system that matches its annual consumption. What is the annual avoided emissions? (Include solar lifecycle EF in the net calculation.)

### Question

**16.** A small island in Lake Victoria relies on a diesel microgrid (EF 0.85). Average daily demand is 800 kWh. A 100 kWp solar PV system with battery is added, supplying 40% of daily energy. Calculate the new daily emissions and the reduction.

### Question

**17.** A cement factory is connected to the grid but also operates a 10 MW heavy fuel oil generator for backup (EF 0.78). If the generator runs 500 hours/year, what is its annual footprint? What percentage of the factory's total carbon footprint (assuming grid main supply 50 GWh/year with EF 0.531) does this represent?

### Question

**18.** A mining company replaces 1.5 MW of diesel generation with hydropower from a nearby plant (EF 0.01). The diesel genset had an average load factor of 0.9 and ran 24/7. Calculate annual emission savings.

### Question

**19.** An office building uses 120 000 kWh/year from the grid (EF 0.531). It signs a VPPA with a wind farm that covers 100% of its consumption. Compute the annual carbon footprint of the building after the VPPA, assuming the wind farm's lifecycle EF is used.

### Question

**20.** A city bus depot installs rooftop solar that generates 300 MWh/year. The depot's annual grid consumption was originally 600 MWh (EF 0.531). Calculate the net carbon footprint reduction, treating solar EF as 0.04.

## Part 2: Energy Intensity (EI) for Electricity (Questions 21–28)

### Question

**21.** Energy intensity (EI) of an economy is defined as energy consumption per GDP (e.g., MJ/USD). If Tanzania's annual electricity consumption is 8 000 GWh and GDP is 75 billion USD, compute the electricity intensity in MJ/USD and kWh/USD.

### Question

**22.** A manufacturing plant produces steel with an energy intensity of 18 GJ/tonne of steel. If the plant produces 200 tonnes per day, what is its daily electricity demand (assuming 70% of energy is electricity)?

### Question

**23.** The energy intensity of a commercial building is 250 kWh/m<sup>2</sup>/year. The building has a floor area of 5000 m<sup>2</sup>. Calculate its annual electricity consumption. If the grid EF is 0.531, what is its annual carbon footprint?

### Question

**24.** A country reduces its electricity intensity from 4.2 MJ/USD to 3.0 MJ/USD while GDP grows from 50 billion to 60 billion USD. By how many GWh does total electricity consumption change? (Assume 1kWh = 3.6MJ).

### Question

**25.** A data centre has a Power Usage Effectiveness (PUE) of 1.8. The IT equipment consumes 500 kW continuously. Calculate the total annual electricity use (including cooling, lighting). If the data centre improves PUE to 1.3, what is the annual energy saving in MWh?

### Question

**26.** A university lab's energy intensity is 80 kWh/m<sup>2</sup>/year. The lab area is 1200 m<sup>2</sup>. After retrofitting LED lighting and efficient HVAC, the intensity drops to 55 kWh/m<sup>2</sup>/year. Compute the annual electricity saved (kWh) and the associated carbon saving (grid EF 0.531).

#### Question

**27.** The average energy intensity of a Tanzanian household is 0.6 kWh per person per day. For a household of 5 persons, calculate monthly consumption. If the national target reduces intensity by 20%, what is the new monthly consumption per household?

#### Question

**28.** An industrial motor consumes 150 kW with an efficiency of 85%. After replacement, the efficiency rises to 94% for the same mechanical output. Calculate the annual energy saving (operating 6000 hours/year) in kWh and the reduction in energy intensity per unit output (assuming constant output).

### Part 3: Water Intensity for Production of Electrical Equipment (Questions 29–36)

#### Question

**29.** The water intensity for manufacturing monocrystalline silicon solar panels is 0.6 L/W<sub>p</sub>. A 5 kWp residential system is installed. How many litres of water are consumed during panel production? (Assume the system uses only panels from this process.)

#### Question

**30.** A PV inverter with a mass of 15 kg is manufactured with a water intensity of 100 L/kg of device. Calculate the total water footprint for producing the inverter. If the inverter lasts 10 years, what is the annualised water footprint?

#### Question

**31.** A large solar farm of 50 MWp uses panels with water intensity 0.55 L/W<sub>p</sub> and mounting structures that add another 20% water consumption per W<sub>p</sub>. Determine the total water volume (m<sup>3</sup>) required for module production.

#### Question

**32.** The production of lithium-ion batteries for energy storage has a water intensity of 15 L/kWh of battery capacity. A home battery system stores 10 kWh. Compute the water consumed for its manufacturing. Compare this to the water used for producing a 3 kWp solar array (solar WI 0.6 L/W<sub>p</sub>).

#### Question

**33.** A wind turbine (2MW) requires 100 tonnes of electronics per MW (inverters, converters). Assuming water intensity 120 L/kg for electronics, what is the water footprint for the electronic components?

#### Question

**34.** A semiconductor fabrication plant produces chips for solar inverters. The water intensity per chip is 8 L/chip. If each inverter contains 200 chips, and the plant produces 10 000 inverters/year, what is the annual water consumption?

#### Question

**35.** A recycling plant recovers 80% of water used in PV module production. If the gross water requirement for a 100 MW<sub>p</sub> factory is 0.6 L/W<sub>p</sub> without recycling, what is the net water consumption after applying 80% recycling? Express in cubic metres.

#### Question

**36.** A country plans to install 500 MW<sub>p</sub> of solar PV. Using a water intensity of 0.65 L/W<sub>p</sub> (including glass and frame), how many Olympic swimming pools (2500 m<sup>3</sup>) of water are needed for manufacturing?

## Part 4: Economic Cost of Air Pollution (PM<sub>2.5</sub>, PM<sub>10</sub>) and Mortality (Questions 37–50)

#### Question

**37.** A diesel generator set operating in a residential area emits 0.5 g of PM<sub>2.5</sub> per kWh. The generator runs 6 hours/day at 20 kW average load. Calculate the daily PM<sub>2.5</sub> emission in kg. If the mortality cost per kg PM<sub>2.5</sub> is 1.2 million TZS, what is the daily health damage cost?

#### Question

**38.** A factory near a city emits 10 tonnes of PM<sub>2.5</sub> annually. Studies show that each tonne of PM<sub>2.5</sub> causes 0.8 premature deaths. Using a statistical value of life (VSL) of 600 million TZS, estimate the total mortality cost per year from the factory's PM<sub>2.5</sub>.

#### Question

**39.** A residential neighbourhood has a background PM<sub>2.5</sub> concentration of 25 µg/m<sup>3</sup> due to traffic and small industries. A new diesel power plant adds an extra 12 µg/m<sup>3</sup> over a population of 50 000 people. The exposure-response function indicates that each 1 µg/m<sup>3</sup> increase raises annual mortality risk by 0.5% above baseline. If baseline mortality is 8 per 1000 per year, calculate excess deaths per year.

#### Question

**40.** Tanzania's health statistics attribute 4000 premature deaths annually to PM<sub>2.5</sub> from generators and industrial sources. If the average VSL is 500 million TZS, what is the total economic loss? Express in billion TZS.

### Question

41. A small workshop uses a 50 kVA diesel genset (load 30 kW) for 10 hours/day, 300 days/year. Emission factor:  $\text{PM}_{2.5} = 0.4 \text{ g/kWh}$ . The cost per kg  $\text{PM}_{2.5}$  (health) is 0.9 million TZS. Compute the annual health cost attributable to the genset.

### Question

42. A cement plant emits 800 tonnes/year of  $\text{PM}_{10}$  (with 40% being  $\text{PM}_{2.5}$ ). The  $\text{PM}_{2.5}$ -only mortality cost factor is 1.1 million TZS per kg  $\text{PM}_{2.5}$ . Calculate the annual mortality cost from the plant's  $\text{PM}_{2.5}$  emissions.

### Question

43. A city plans to replace 200 old diesel generators (each average 15 kW, run 4 h/day, 200 days/year) with grid electricity. Each generator emits 0.45 g/kWh  $\text{PM}_{2.5}$ . What is the total annual reduction in  $\text{PM}_{2.5}$  emissions (kg) and the associated health cost saving (use 1.0 million TZS/kg  $\text{PM}_{2.5}$ )?

### Question

44. An industrial area near a residential zone has  $\text{PM}_{2.5}$  levels of  $35 \mu\text{g}/\text{m}^3$  above the WHO guideline ( $5 \mu\text{g}/\text{m}^3$ ). The exposed population is 200 000. The concentration-response coefficient is 0.06 per  $10 \mu\text{g}/\text{m}^3$  increase in all-cause mortality. Baseline mortality rate is 7/1000 per year. Calculate the estimated excess deaths per year due to the pollution.

### Question

45. A remote village uses a single 100 kW diesel generator, 24/7, emitting  $\text{PM}_{2.5}$  at  $0.6 \text{ g/kWh}$ . The village population is 1200. Using a simple box model, the incremental  $\text{PM}_{2.5}$  concentration is estimated at  $8 \mu\text{g}/\text{m}^3$ . The annual mortality cost per  $\mu\text{g}/\text{m}^3$  per person is 50 000 TZS. Compute the total annual health cost.

### Question

46. A large hospital installs a backup diesel generator (1 MW). It runs only during outages, totalling 200 hours/year at half load (500 kW). Emission factor  $\text{PM}_{2.5} = 0.3 \text{ g/kWh}$ . What is the annual  $\text{PM}_{2.5}$  emission? If the hospital is in a densely populated area, the damage cost is 1.8 million TZS/kg  $\text{PM}_{2.5}$ . Find the damage cost.

### Question

47. A policy aims to reduce  $\text{PM}_{2.5}$  from industrial boilers by 30% nationwide. Current  $\text{PM}_{2.5}$  emissions from industry are 15 000 tonnes/year. The health cost per tonne  $\text{PM}_{2.5}$  is 1.4 billion TZS. Calculate the annual health benefit (in billion TZS) after achieving the reduction.



### Question

**48.** A mining operation uses heavy diesel trucks and generators. The total annual PM<sub>2.5</sub> emission is 120 tonnes. Of this, 40% affects a nearby town of 30 000 people. If the mortality cost per kg PM<sub>2.5</sub> is 0.95 million TZS, what is the economic impact on the town?

### Question

**49.** A factory replaces an old diesel generator (PM<sub>2.5</sub> EF 0.7 g/kWh) with a grid connection (grid electricity is cleaner but still has upstream emissions; assume no local PM<sub>2.5</sub> from grid). The generator previously ran 4000 hours/year at 100 kW average. How many kg of PM<sub>2.5</sub> are avoided each year? What is the health cost saving (use 1.0 million TZS/kg)?

### Question

**50.** A combined heat and power (CHP) plant running on natural gas emits PM<sub>2.5</sub> at 0.05 g/kWh. It serves a district heating system. If the plant produces 20 GWh/year, calculate its total PM<sub>2.5</sub> emission (kg). Using a damage factor of 0.8 million TZS/kg, determine the health cost. How much lower is this compared to a diesel plant with EF 0.5 g/kWh for the same output?